

# Arpan Pathak

📍 Seattle, Washington, USA | Open to Relocation

✉ Email: arpan.pathak47@gmail.com | Phone: +1 (206) 306-6059

🌐 LinkedIn: [linkedin.com/in/arpan-pathak-272341424](https://www.linkedin.com/in/arpan-pathak-272341424) | GitHub: [github.com/arpanpathak](https://github.com/arpanpathak)

## 📄 SUMMARY

Senior Software Engineer, **8 years** building developer tools, observability platforms, and performance-critical distributed systems at **Amazon, Microsoft, Oracle**. Full-stack **Rust, C++, Kotlin, Go, Python, JavaScript**; **Kubernetes, eBPF/Cilium**, GPU/CUDA, cloud **performance optimization**.

## 🏢 EXPERIENCE

### Software Developer II, Microsoft, Redmond, WA

Jun 2025 - Aug 2026

- **Zero-Trust Sovereign Cloud Networking (eBPF/Cilium)**: Architected zero-trust sovereign cloud networking with **eBPF + Cilium** network policies enforcing **data residency** and **SecNumCloud** compliance; authored **Kubernetes operators (Kubebuilder)** to reconcile policy state across AKS clusters in France and Germany.
  - **C++/C# to Rust Migration (Memory Safety)**: Spearheaded migration of the Information Protection service from **unsafe C++** and **C#** to **Rust**, moving memory safety to compile time: eliminated use-after-free, buffer-overflow, and null-safety bugs in the migrated path. Delivered ~ **40% lower P99 latency**, **2x throughput**, **3x lower memory** per request, and **6-figure annual cost savings**.
  - **AI-Powered Policy Engine**: Built an AI engine that auto-generates Cilium policies and scans for misconfigurations: manual review cut from **4 hours** to **near-real-time**, zero policy drift across sovereign regions.
- 🔗 TechStack: **Rust, C++, eBPF, Cilium, Kubernetes (AKS), Kubebuilder, Prometheus, Linux**

### Senior Software Engineer, Oracle Cloud Infrastructure, Seattle, WA

Oct 2024 - Mar 2025

- **High-Performance Database Engine**: Architected low-latency internals for a proprietary database engine, optimizing **lock-free data structures** to reduce contention in high-concurrency read-write paths, improving transaction throughput by **45,000 ops/s**.
  - **Infrastructure-as-Code SDK**: Engineered a **Terraform provider** (developer-facing SDK) for Oracle Autonomous Database, replacing legacy control plane infrastructure with **\$1.2M/month** projected savings.
  - **Security & Key Management**: Integrated OCI Security Vault across control plane services for secure secret management and automated key rotation protocols.
- 🔗 TechStack: **Java, Go, Terraform, Oracle Cloud Infrastructure, Autonomous Database**

### Software Development Engineer II, Amazon, Seattle, WA & Hyderabad, India

Mar 2021 - Oct 2024

- **Real-Time ML Ranking Engine**: Architected an **XGBoost + Deep Learning ensemble** for real-time ads & deal ranking across FreeVee, Twitch, MiniTV, Prime Video, and Amazon Retail, driving **5% CTR increase**, **30%** higher deal impressions, and **\$5.4M monthly revenue growth**.
  - **BERT Inference Service**: Delivered **BERT-based** NLP inference handling **5,000 QPS** at peak, cutting support-ticket triage SLA from **7 days** to **2 hours**.
  - **Event Bus SDK (Kotlin, CSP)**: Built an event bus SDK using **Kotlin structured concurrency** and the **CSP (communicating sequential processes)** paradigm to deliver programmatic guaranteed ad platform deals in real-time with **< 50ms end-to-end latency**.
  - **Distributed Systems & SDK Design**: Designed uniqueness-constraint indexing and an **SDK** for a multi-tenant NoSQL store using **two-phase commit**, achieving **serializable isolation** at **3K writes/second**.
  - **Big Data Platform**: Built big data infrastructure & pipelines for the Amazon WorkEvents data platform stream team: **1-petabyte** data warehouse & data lake serving **1M+ daily BI queries**; wrote business logic in **AWS Glue (serverless Apache Spark)** for financial reporting and pay computation.
- 🔗 TechStack: **Kotlin, Java, Python, PyTorch, XGBoost, Spark, Kafka, DynamoDB, Redshift, AWS**

### Software Development Engineer, Razorpay, Bengaluru, India

Aug 2020 - Feb 2021

- Built UPI Payment Gateway Aggregator in **Go** + Protobuf processing **1M+ hourly transactions** with **99.99% uptime** at **150ms P99** under **50,000 TPM** peak load.
- 🔗 TechStack: **Go, Redis, PostgreSQL, Protobuf, Kafka**

### Software Engineer, Mindfire Solutions, Bhubaneswar, India

Aug 2018 - Feb 2020

- Delivered full-stack workflow tools and an **AR-based product search microservice** serving **10,000+ daily queries**; cut dashboard latency from **5 min to 1 min**.

<> **TechStack:** *Java, Spring Boot, Angular, MySQL, MongoDB*

## 🎯 PERSONAL PROJECTS

### 📦 **Driving-CivicSense: Edge AI Vision System (Rust)**

- Real-time on-device perception pipeline in **Rust** on the **NVIDIA Jetson Orin Nano Super** (67 INT8 TOPS, 8 GB, 7-15 W): **YOLOv8/v11 INT8 ONNX** (~ **12 ms**) + Deep SORT/Kalman tracking + formally-proven kinematic decision engine; **100%** on-device, privacy-first.
- Research paper: **Deterministic Intersection Blockage Prediction** | [arpanpathak.github.io/driving-civicsense-vision-model/](https://arpanpathak.github.io/driving-civicsense-vision-model/) | Code: [github.com/arpanpathak/driving-civicsense-vision-model](https://github.com/arpanpathak/driving-civicsense-vision-model)
- Books: **Seeing Machines** | [arpanpathak.github.io/seeing-machines-book](https://arpanpathak.github.io/seeing-machines-book) · **CUDA Kernels** (CUDA-Oxide / Rust-to-CUDA) | [arpanpathak.github.io/gpu-parallel-book](https://arpanpathak.github.io/gpu-parallel-book)

## 🔧 SKILLS

<> **Languages:** Rust, C++, Kotlin, Go, C, Java, Python, TypeScript, JavaScript, Shell Script

🔧 **Systems Programming & Performance:** Lock-Free Data Structures, Concurrency, Memory Safety, Memory Profiling, Cache Optimization, Performance Engineering, Low-Latency Systems, OS Internals, Computer Architecture

🌐 **Networking & Protocols:** TCP/IP, UDP, TLS, mTLS, DNS, HTTP/2, HTTP/3, QUIC, WebSockets, RDMA, InfiniBand, DPDK, VXLAN, GRE, Socket Programming, Packet Processing, Network Performance Optimization

🔒 **Network & Cloud Security:** eBPF, XDP, Cilium, Kernel-Level Networking, Packet Processing, VPC, Subnets, Routing, NAT, VPN, Peering, Load Balancing, Security Groups / Firewalls

📁 **Distributed Systems & Streaming:** Distributed Systems (consensus, replication, sharding), Apache Kafka, Apache Flink, Apache Spark, Spark Streaming, AWS Kinesis, Real-Time Data Pipelines, Change Data Capture (CDC), Microservices

🔗 **APIs & Serialization:** gRPC, REST API Design, Protobuf

⚡ **ML Inference & AI:** Deep Learning, Computer Vision, PyTorch, TensorFlow, XGBoost, Transformer Models (BERT), LLMs, ONNX Runtime, On-Device/Edge Inference, Model Pruning & Quantization, YOLOv8/v11, CNN, NLP, Recommendation Systems, MLOps

🔧 **GPU & Parallel Computing:** CUDA, CUDA-Oxide (Rust-to-CUDA), SIMT, GPU Kernel Programming, Parallel Computing, GPU-Accelerated Computing

☁ **Cloud & Infra:** AWS (EC2, ECS, Lambda, Kinesis, EMR, SageMaker), Azure (AKS), Google Cloud, Kubernetes (EKS, AKS), Docker, Terraform, Kubebuilder, Istio, Kubernetes Gateway API

📊 **Observability:** Prometheus, Grafana, Azure Monitor, AWS CloudWatch

🗄 **Data & Storage:** DynamoDB, PostgreSQL, Redis, MongoDB, MySQL, Amazon Redshift, Amazon S3

🔗 **Core CS:** Data Structures & Algorithms, System Design, Operating Systems, Linux

## 🎓 EDUCATION

**B.Tech, Computer Science & Engineering**, RCC Institute of Information Technology, Kolkata | 2018